

✓ PDS PRACTICAL 3

NAME : DIVYA R. WAGHMODE

DIV : SY-A

ROLL NUMBER: A-28

Aim = Data Wrangling with pandas: Manipulate and transform data to suit the analysis needs.

1 . What is data wrangling? Explain the difference between data wrangling and data preprocessing.

Data Wrangling Data wrangling means cleaning, organizing, and transforming raw data into a useful format for analysis.

It includes tasks like:

- Removing missing values
- Correcting wrong data
- Changing data types
- Renaming columns
- Combining multiple datasets
- Formatting data properly

Difference between Data Wrangling and Data Preprocessing

Data Wrangling:

- Focuses on cleaning and organizing raw data
- Includes handling missing values, duplicates, formatting

Data Preprocessing:

- Focuses on preparing data for machine learning or analysis
- Includes normalization, scaling, encoding

2 . What is the importance of exploratory data analysis (EDA) in Pandas? EDA means **Exploratory Data Analysis**. It is used to understand the data before performing analysis or building models.

Importance of EDA in Pandas: -Helps to understand the structure of data

- Finds missing values
- Detects outliers
- Checks data types
- Shows patterns and relationships
- Helps in better decision making
- Improves data quality

Common Pandas functions used in EDA

`head()`

`tail()`

`info()`

`describe()`

3.What are the main differences between Series, DataFrame, and Panel?

1.Series

- One-dimensional data structure
- Like a single column
- Stores values with index

Example:

```
import pandas as pd
```

```
s = pd.Series([10, 20, 30])
```

2.DataFrame

- Two-dimensional data structure
- Rows and columns
- Like a table or Excel sheet

Example:

```
df = pd.DataFrame( { 'Name': ['A', 'B'], 'Marks': [80, 90] })
```

3.Panel

- Three-dimensional data structure
- Collection of multiple DataFrames

4 . Explain the use of:

a. `isnull()`

b. `notnull()`

c. dropna()

d. fillna()

1 . isnull()

Used to check missing (null) values in data.

Returns True for missing values.

Example:

```
df.isnull()
```

b) notnull()

-Used to check non-missing values.

-Returns True where value is present.

Example:

```
df.notnull()
```

c) dropna()

-Used to remove rows or columns containing missing values.

Example:

```
df.dropna()
```

d) fillna()

-Used to replace missing values with some value.

Example:

```
df.fillna(0)
```

5 . What is column transformation and why is it important in data analysis?

Column Transformation

Column transformation means changing the values or format of a column to make it useful for analysis.

Examples:

-Converting text to numbers

-Changing uppercase to lowercase

-Scaling values

-Creating new columns from old ones

Example:

```
df['Age'] = df['Age'] + 1
```

```
import pandas as pd
```

```
df = pd.read_csv("/content/train (1).csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 211
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs)	female	38.0	1	0	PC 175

```
age_column = df["Age"]
print(age_column.head())
```

```
0    22.0
1    38.0
2    26.0
3    35.0
4    35.0
Name: Age, dtype: float64
```

```
df["Source"] = "Titanic Dataset"
print(df[["Name", "Source"]].head())
```

	Name	Source
0	Braund, Mr. Owen Harris	Titanic Dataset
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	Titanic Dataset
2	Heikkinen, Miss. Laina	Titanic Dataset
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	Titanic Dataset
4	Allen, Mr. William Henry	Titanic Dataset

```
sorted_df = df.sort_values(by="Fare", ascending=False)
print(sorted_df[["Name", "Fare"]].head())
```

	Name	Fare
679	Cardeza, Mr. Thomas Drake Martinez	512.3292
258	Ward, Miss. Anna	512.3292
737	Lesurer, Mr. Gustave J	512.3292
88	Fortune, Miss. Mabel Helen	263.0000
438	Fortune, Mr. Mark	263.0000

```
df_renamed = df.rename(columns={
    "Pclass": "Passenger_Class",
    "SibSp": "Siblings_Spouses",
    "Parch": "Parents_Children"
})
print(df_renamed.columns.tolist())
```

```
['PassengerId', 'Survived', 'Passenger_Class', 'Name', 'Sex', 'Age', 'Sibling
```

```
print(df.describe())
```

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
grouped = df.groupby("Pclass")[["Fare", "Age"]].mean()
print(grouped)
```

Pclass	Fare	Age
1	84.154687	38.233441
2	20.662183	29.877630
3	13.675550	25.140620

```
extra_info = pd.DataFrame({
    "PassengerId": [1, 2, 3, 4, 5],
    "Nationality": ["British", "American", "Irish", "French", "German"]
})
```

```
merged_df = pd.merge(df, extra_info, on="PassengerId", how="left")
print(merged_df[["PassengerId", "Name", "Nationality"]].head())
```

	PassengerId	Name	Nationality
0	1	Braund, Mr. Owen Harris	British
1	2	Cumings, Mrs. John Bradley (Florence Briggs Th...	American
2	3	Heikkinen, Miss. Laina	Irish
3	4	Futrelle, Mrs. Jacques Heath (Lily May Peel)	French
4	5	Allen, Mr. William Henry	German

```
df_dropped_cols = df.drop(columns=["Cabin", "Ticket", "Source"])
print("After dropping columns:", df_dropped_cols.shape)
```

After dropping columns: (891, 10)

```
df_dropped_rows = df_dropped_cols.dropna(subset=["Age"])
print("After dropping rows with missing Age:", df_dropped_rows.shape)
```

After dropping rows with missing Age: (714, 10)

```
def age_group(age):
    if age < 18:
        return "Child"
    elif age < 60:
        return "Adult"
    else:
        return "Senior"

df["Age_Group"] = df["Age"].apply(age_group)
print(df[["Name", "Age", "Age_Group"]].head(10))
```

	Name	Age	Age_Group
0	Braund, Mr. Owen Harris	22.0	Adult
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	Adult
2	Heikkinen, Miss. Laina	26.0	Adult
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	Adult
4	Allen, Mr. William Henry	35.0	Adult
5	Moran, Mr. James	NaN	Senior
6	McCarthy, Mr. Timothy J	54.0	Adult
7	Palsson, Master. Gosta Leonard	2.0	Child
8	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	27.0	Adult
9	Nasser, Mrs. Nicholas (Adele Achem)	14.0	Child

conclusion

In this practical, we performed Data Wrangling using the pandas library in Python on the Titanic dataset (train.csv) containing 891 rows and 12 columns such as PassengerId, Survived, Pclass, Age, Fare, Sex, and more. Basic Operations (Tasks 1–5) covered the foundational skills of pandas — selecting columns, adding new columns with constant values, sorting data, renaming columns, and generating summary statistics using describe(). These operations help in understanding and organizing the raw dataset. Moderate Operations (Tasks 6–10) involved more analytical tasks — filtering rows using multiple conditions (e.g., female passengers who survived), grouping data by passenger class to find average Fare and Age, merging two DataFrames on a common key, dropping unwanted rows and columns to clean the data, and applying custom functions to create new derived columns like Age Group (Child / Adult / Senior). Through these 10 tasks, we learned how pandas provides powerful and flexible tools to clean, transform, filter, and analyze structured data efficiently. These operations form the core of any real-world data analysis or machine learning pipeline, making pandas an essential skill for any Data Science practitioner.

